The best model (40_2_mean) shows a relatively tight error distribution, with **MSE values mostly below 175** and **MAE centered around 0.5**, indicating consistent and accurate predictions.

Enhancing Prognostic Accuracy in AKI: A Systematic Data Quality Assurance Framework for CRRT Machine Data

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Summary

Objective: Improve the quality and integrity of CRRT machine time-series data to support mortality prediction in AKI patients

Data Source and Patient Cohort:

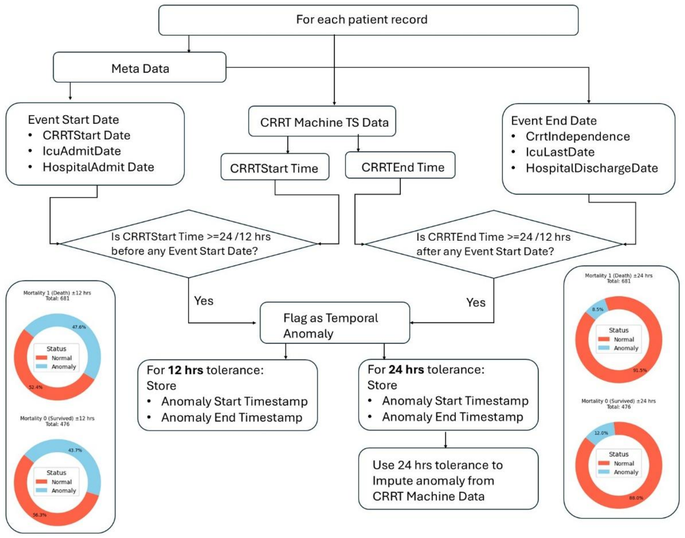
- CRRTnet ,registry of CRRT treatment data along with patient outcome for critically ill adults (aged ≥18) (from Dec. 2013 to Sep. 2020) is used in this study.
- Pressure features (ACCESS, FILTER, RETURN, EFFLUENT) are used as Time Series data stream
- A total of 1157 patients were included in the analysis. The median age of the survivors (count: 476) was 57 years (IQR: 48–67), compared to 62 years (IQR: 53–71) for those who died (count: 681), with 41.93% women in the cohort.

Methods: Designed a multi-step data **quality assurance pipeline**:

- Temporal anomaly detection using clinical timestamp inconsistencies
- Removal of machine prime and inactive sessions
- Outlier injection and detection simulation using synthetic anomaly
- Developed a rolling-window based smoothing algorithm using customized parameters
- Evaluated the data quality assurance framework using tunable parameters with Precision, Recall, F1 Score, MSE and MAE.

Methods

Step 1: Temporal Anomaly Detection and Imputation



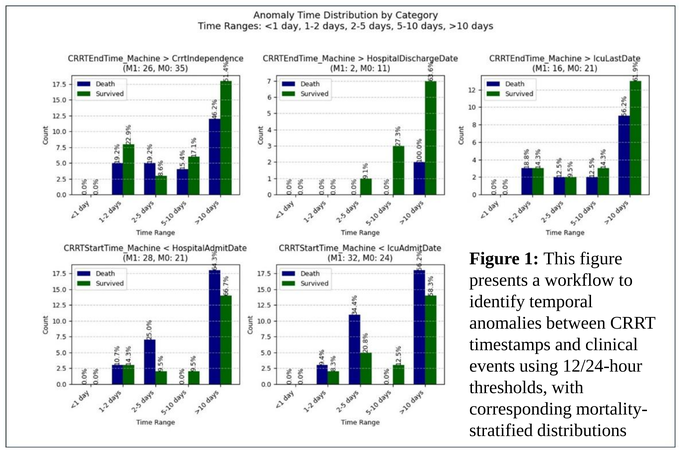


Figure 1: This figure presents a workflow to identify temporal anomalies between CRRT timestamps and clinical events using 12/24-hour thresholds, with corresponding mortality-stratified distributions

Methods

Step 2: Removal of treatment inactive sessions

- **Prime Session**: This is the preparatory phase where the CRRT circuit is flushed with a priming solution (e.g., saline) to eliminate air bubbles and ensure the system is safe for blood flow before connecting to the patient.
- **Inactive Session**: Refers to periods when the machine is on but not delivering therapy—commonly during setup, troubleshooting, pauses in treatment, or patient transport.

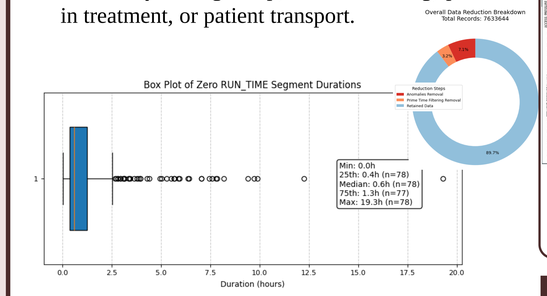


Figure: 2: The durations with RUN_TIME = 0.0 (ranging from 0.0 h to 19.3 h, median = 0.6 h) likely capture periods when the CRRT machine was in prime mode or inactive sessions between treatments. Short durations (≤ 1 h) often reflect priming or temporary pauses, while longer outliers (e.g., > 5 h) suggest extended inactivity, possibly due to patient transfer, troubleshooting, or therapy discontinuation. This characterization helps differentiate brief operational gaps from clinically significant idle times.

Step 3: Outlier Injection, Detection, and Smoothing Workflow on CRRT TS Data

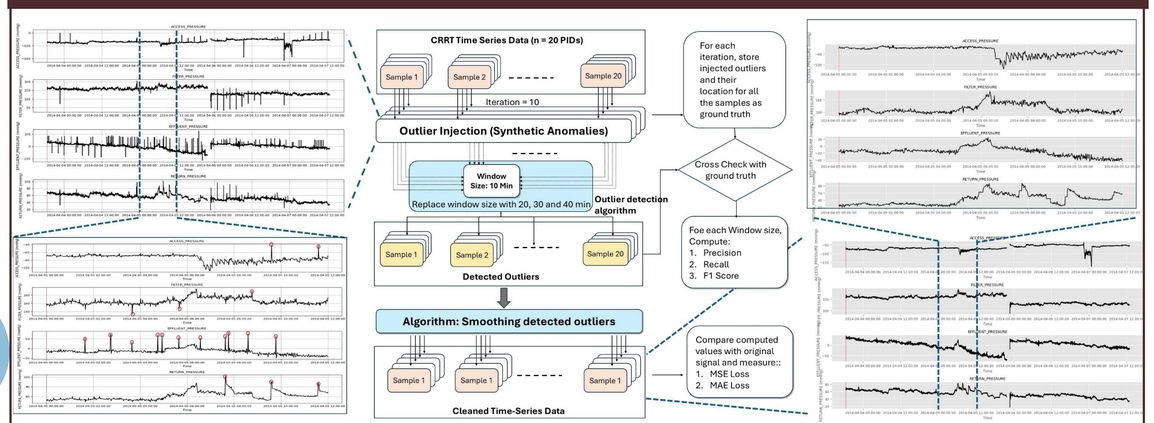


Figure 3: This figure demonstrates the anomaly detection and correction pipeline applied to CRRT time-series data from ***n*= 20 patient samples**. For each of the **10 iterations**, synthetic outliers were injected using a randomly selected noise factor $f \in [-5,5]$ The left panel illustrates pressure signals with injected anomalies, while the right panel presents the corresponding smoothed outputs after applying the detection and correction algorithms.

Outlier Detection and Smoothing Algorithm

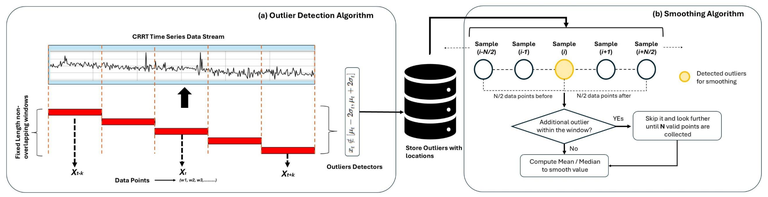


Figure 4 : Pressure signal values X_t are flagged as outliers if $x_t \notin [\mu_t - 2\sigma_t, \mu_t + 2\sigma_t]$ within a fixed window. Each outlier is then smoothed using the mean or median of N surrounding non-outlier points, where the set dynamically excludes additional anomalies to maintain exactly N valid values.

Results

Table 1: Best Outlier detection model based on window size

Window size : [10, 20, 30, 40]
Data Points : [2, 6, 10, 12]
Smoothing method : [‘mean’, ‘median’]

Model	Window Size	Data Points	Smoothing method	Precision	Recall	F1 scores
1	10	2	mean	0.594	0.642	0.627
2	20	2	median	0.647	0.892	0.75
3	30	2	mean	0.718	0.929	0.809
4	**40**	**2**	**mean**	**0.778**	**0.937**	**0.85**

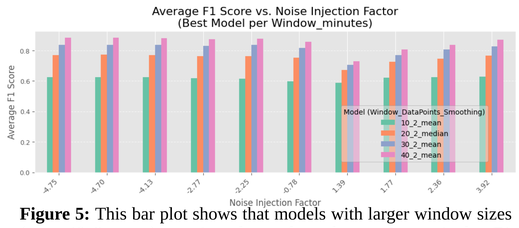


Figure 5: This bar plot shows that models with larger window sizes (e.g., 40_2_mean) consistently perform better. Interestingly, F1 scores are lowest near zero noise (e.g., -0.78, 1.39) and increase as noise becomes more extreme

Table 2: Top 5 Smoothing models with best window size (40)

Model	Window Size	Data Points	Smoothing method	F1 scores	MSE	MAE
1	**40**	**2**	**mean**	**0.85**	**83.58**	**0.521**
2	40	2	median	0.8489	157.803	0.603
3	40	12	mean	0.8486	175.479	0.663
4	40	6	mean	0.8485	160.782	0.578
5	40	12	median	0.8482	167.283	0.602

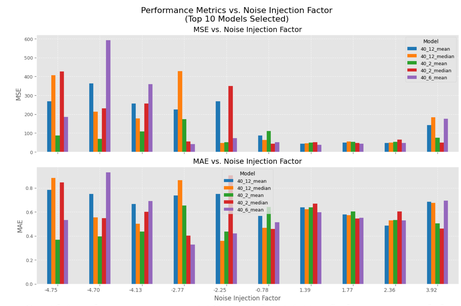


Figure 6: This plot compares MSE and MAE of the top models across various noise injection factors, where lower values indicate better performance.



Figure 7: The best model (40_2_mean) shows a relatively tight error distribution, with MSE values mostly below 175 and MAE centered around 0.5, indicating consistent and accurate predictions.

Conclusions

- Among tested configurations, *40_2_mean* delivered the highest F1 score (0.85) while maintaining low MSE (83.58) and MAE (0.521), demonstrating optimal balance between outlier detection and signal restoration.
- We aim to apply long sequence modeling for rolling mortality prediction, leveraging our framework to ensure the required high-integrity data streams for accurate and timely clinical decision-making in CRRT care.

Acknowledgement

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